SHP ASSET MANAGER

CABLE TRIGGER

User Manual

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SECTION 01

HARDWARE OVERVIEW

The **Cable Trigger** is a servo-driven shutter release that physically actuates your camera's cable-release port. A small servo arm presses the plunger to a position you dial in through the app — no radio frequency tricks, no electronic handshakes — just a mechanical press of the cable release your camera already knows and trusts.

Because it works at the mechanical level it is compatible with cameras that do not expose an electronic remote-trigger port, and it produces the same signal your camera's built-in shutter button produces.

WHAT THE APP CONTROLS

PARAMETER	WHAT IT SETS
Pre-Trigger Delay	Time between the app sending the trigger command and the servo moving. Use to give yourself time to step away before the shot fires, or to synchronise with another event.
Exposure Time	How long the servo holds the cable release depressed. Match this to your camera's bulb-exposure requirement, or the minimum dwell time the shutter needs to register a press.
Trigger Engagement Amount	The servo angle (expressed as a percentage of its full range) used during each trigger. Each camera and cable-release combination has a different sweet spot — this dials it in precisely.

CONNECTIVITY

The Cable Trigger communicates with the SHP Asset Manager app over **Bluetooth Low Energy (BLE)** and over a **WiFi hotspot** it creates itself. Both modes provide full, real-time control of all settings.

SYSTEM STATUS

The large status display at the top of the Cable Trigger tab shows the current state of the device and a COUNTDOWN timer. States you will see:

STATE	WHAT THE DEVICE IS DOING
IDLE	Ready and waiting. Tap TRIGGER SEQUENCE to fire.
DELAYING	Pre-trigger delay is counting down before the servo moves.

STATE	WHAT THE DEVICE IS DOING
MOVING OUT	Servo is travelling to the engagement position.
HOLDING	Servo is holding the cable release depressed for the Exposure Time duration.
RETURNING	Servo is returning to the home position after the exposure.

The arc gauge below the status display animates in real time, sweeping from 0% (home) to the set engagement amount as the servo travels.

STATUS LIGHT

The Cable Trigger has a small status LED on the board. It tells you at a glance what the device is doing — useful the moment you power on and before the app has connected.

LED PATTERN	WHAT IT MEANS
Two quick flashes at boot	Firmware is running — the board powered up correctly.
Slow pulse (brief flash every ~2 s)	Advertising over BLE — waiting for the app to connect.
Steady on	Connected and idle — ready to trigger.
Fast blink (on/off every 125 ms)	Trigger active — servo is moving or holding the cable release.

> If the LED is off entirely, the device is in deep sleep. Wake it by pressing its power or wake button, or cycling power.

SECTION 02

CONNECTING TO THE APP

The SHP Asset Manager app controls all Cable Trigger settings in real time. Connect via **Bluetooth** or **WiFi** — both give full control. WiFi is recommended for firmware updates due to significantly faster transfer speeds.

BLUETOOTH

1. Power on the Cable Trigger.
2. Ensure Bluetooth is enabled on your phone.
3. Open the app and tap the **CABLE TRIGGER** tab at the bottom of the screen.
4. Tap ⚡ **CONNECT DEVICE** in the header and select your device from the scan list.
5. Settings sync automatically. The dot in the header turns orange when connected.

> The app remembers your device. On subsequent launches it reconnects silently if the Cable Trigger is in range. Tapping CONNECT DEVICE when a device is already saved gives you the option to reconnect to it, scan for a new device, or connect via WiFi.

WIFI

The Cable Trigger creates its own WiFi hotspot. WiFi is recommended when installing firmware updates — it is significantly faster than Bluetooth.

1. Power on the Cable Trigger.
2. On your phone, join the Cable Trigger's network in your OS WiFi settings. Its hotspot password is **12345678**; the app Security PIN does not change this password.
3. Return to the app. Tap ⚡ **CONNECT DEVICE** → *Connect via WiFi*.
4. Enter IP **192.168.4.1** and port **80**, then tap CONNECT.
5. Settings sync automatically once connected.

USING CABLE TRIGGER WITH ADVANCED TX

Advanced TX is the Raspberry Pi camera/control unit. Its PI TX tab and browser panel control the Pi's own camera, trigger output, wake output, and paired RX units; they do not operate the Cable Trigger's servo or change its settings. Connect to the Cable Trigger in the Cable Trigger tab for those controls. To have Lazer Eye detections fire both devices, pair the Lazer Eye TX with the Cable Trigger using the **DIRECT CABLE**

TRIGGER LINK, and separately link the Lazer Eye TX to Advanced TX using the **ADVANCED TX LINK**. See the Lazer Eye manual, Section 07 for RX and Advanced TX pairing, and Section 08 for the Cable Trigger link.

> The Advanced TX browser panel requires a trusted network. It does not require login, so do not expose it to a public or untrusted network.

SETTINGS

Tap the ⚙ **gear icon** in the header to open Settings. From there, Hardware Settings includes FIRMWARE UPDATE (OTA), DEVICE NAME, and SECURITY PIN. The Cable Trigger WiFi hotspot uses its fixed password `12345678`; its Security PIN is for BLE authentication, not the WiFi password. See the Lazer Eye manual, Section 02 for the Advanced TX connection and network distinction.

SECTION 03

CONFIGURATION SETTINGS

The **CONFIGURATION** card is the main control panel for the Cable Trigger. Tap its header to expand or collapse it. All values are written to the device immediately when changed and are retained after power-off.

PRE-TRIGGER DELAY

0 s to 18,000 s

Time between the trigger command leaving the app and the servo starting to move. Increase to give yourself time to step back from the camera. Small values step in 1-second increments; above 60 s they step in 1-minute increments; above 1 hour they step in 10-minute increments.

EXPOSURE TIME

0 s to 18,000 s

How long the servo holds the cable release depressed. For a standard shot, set this slightly longer than your camera's minimum registration time. For bulb exposures, set it to the full exposure length. Follows the same step increments as Pre-Trigger Delay. Setting this to 0 still sends a brief minimum pulse to the firmware so the servo completes a full actuation cycle.

The compact summary line at the top of the collapsed CONFIGURATION card shows current values at a glance: `DELAY · EXP · ENG`.

TRIGGERING

Tap the **TRIGGER SEQUENCE** button at the bottom of the status card to fire the shutter immediately. The button is only active when the device is IDLE and connected (or in simulation mode).

SECTION 04

TRIGGER ENGAGEMENT AMOUNT

Every cable release has a pre-travel zone — a short distance the plunger moves before it actually trips the shutter. **Trigger Engagement Amount** sets how far the servo travels (as a percentage of its full range) to engage the release. This is the single control that determines servo rotation depth.

Tap the **TRIGGER ENGAGEMENT AMOUNT** row inside the CONFIGURATION card to expand it. A `ParamControl` slider with +/− buttons lets you set the value from **1% to 100%**.

The arc gauge in the status card animates in real time as you adjust the value, giving live visual feedback of where the servo will travel to.

FINDING THE RIGHT ENGAGEMENT AMOUNT

1. Attach the cable release to your camera and mount the Cable Trigger.
2. Set Pre-Trigger Delay to `0 s` and Exposure Time to a comfortable value (e.g. `0.5 s`).
3. Start with a low Trigger Engagement Amount (e.g. 20%) and tap **TRIGGER SEQUENCE**.
4. Gradually increase the percentage until the shutter fires reliably on every press.
5. Find the minimum % that fires reliably — this avoids over-stressing the release mechanism.

Do not set Trigger Engagement Amount higher than needed. Over-driving a cable release can stress the mechanism and damage the camera's shutter port.

> All settings are written to the device immediately when changed and survive power cycles.

SECTION 05

PRESETS

Presets save a complete trigger configuration — Pre-Trigger Delay, Exposure Time, and Trigger Engagement Amount — under a name you choose. Up to **five presets** can be stored. They are saved on your phone and are not affected by device connections or firmware updates.

The PRESETS row appears at the top of the expanded CONFIGURATION card.

SAVING A PRESET

1. Dial in your settings (Pre-Trigger Delay, Exposure Time, Trigger Engagement Amount) for a particular camera or shooting scenario.
2. In the PRESETS row, tap the dashed **SAVE CURRENT AS PRESET** button, or the **+ SAVE** button if presets already exist.
3. Type a name (up to 16 characters) and tap the ✓ confirm button or press Return.

LOADING A PRESET

Tap a preset chip in the horizontal PRESETS list to load it instantly. All three values (Pre-Trigger Delay, Exposure Time, Trigger Engagement Amount) are applied immediately and synced to the device if connected.

DELETING A PRESET

Tap the × button on the right side of any preset chip to delete it. The slot is freed and a new one can be saved in its place.

> Each preset chip shows a compact summary: `D:` = Pre-Trigger Delay, `E:` = Exposure Time, `R:` = Trigger Engagement Amount %.

SECTION 06

SHUTTER CALIBRATION

Shutter Calibration uses your phone's microphone to record the shutter sound and estimate an Exposure Time offset for a specific camera. It does not change the Cable Trigger automatically.

The SHUTTER CALIBRATION card is a separate collapsible section below the CONFIGURATION card. Tap its header to expand it.

RUNNING A CALIBRATION TEST

1. Dial in Trigger Engagement Amount first — the test fires a real shot, so the servo must already be reaching the shutter.
2. Enter a name for this calibration (e.g. your camera body or lens) in the **LENS / SHUTTER NAME** field at the top of the section.
3. Hold your phone close to the camera (within 15–20 cm), microphone facing the camera body where the shutter sound is loudest.
4. Make sure the area is quiet — ambient noise reduces detection accuracy.
5. Tap **TEST SHUTTER**. The app counts down, fires one shot, and records audio during the cycle.
6. After a brief analysis phase, the EXPECTED / DETECTED / OFFSET metrics appear, with a waveform showing the open and close click peaks.
7. Review the offset. A positive offset means the firmware held longer than the shutter was open; a negative offset means it closed before the hold ended.
8. Tap **SAVE PROFILE** to store the result under the name you entered.

APPLYING A SAVED PROFILE

Saved profiles appear in the **SAVED PROFILES** list at the bottom of the Shutter Calibration section. Each row shows the profile name, the date it was tested, and its offset in milliseconds.

Tap **APPLY** on any profile to send the current Exposure Time plus the saved offset to the device for this connection. The app does not overwrite the stored Exposure Time.

> Applied calibration lasts until you disconnect or change Exposure Time. It is temporary; saved profiles stay on the phone. Apply the profile again when needed.

> Grant microphone permission when prompted. Without it the app cannot record audio and TEST SHUTTER

will fail with a permission error.

WHEN TO RECALIBRATE

- When you switch to a different camera body.
- When you change the cable release mechanism.
- When you find the shutter is consistently held too short or too long after a firmware update.
- When shooting in new temperature conditions (thermal expansion can shift the actuation point slightly).

SECTION 07

REMOTE TRIGGER MODE

Remote Trigger Mode lets you fire the camera without tapping the on-screen TRIGGER SEQUENCE button. The toggle is at the bottom of the CONFIGURATION card, below the Longevity Test row.

> The screen stays on automatically while Remote Trigger Mode is active, so the phone does not sleep between shots. All shot settings (Pre-Trigger Delay, Exposure Time, Trigger Engagement Amount) still apply when firing remotely.

ANDROID — VOLUME BUTTONS

While Remote Trigger Mode is on, pressing the physical **Volume Up** or **Volume Down** buttons on your Android phone fires the shutter. This lets you hold the phone in one hand and trigger with a thumb, keeping your other hand free to steady the camera.

IOS — BLUETOOTH CLICKER

On iOS, pair a Bluetooth shutter remote (a BLE clicker, such as the kind sold for selfies or remote presentations) with your phone. While Remote Trigger Mode is on, pressing the clicker fires the Cable Trigger. The clicker can be held in one hand or placed on a surface nearby.

WEB — KEYBOARD

When using the web app, keep the Cable Trigger page active and press **Space**, **Enter**, or **Up Arrow** to fire. The web keyboard shortcut is only available while the page is active.

PLATFORM	REMOTE INPUT	HOW TO ACTIVATE
Android	Volume Up or Volume Down (hardware buttons)	Enable the Remote Trigger Mode toggle inside the CONFIGURATION card. No additional hardware required.
iOS	BLE shutter clicker (paired in iOS Bluetooth settings)	Pair the clicker via iOS Settings → Bluetooth, then enable the Remote Trigger Mode toggle in the app.
Web	Space, Enter, or Up Arrow	Keep the Cable Trigger page active and enable Remote Trigger Mode.

> Disable Remote Trigger Mode when you are done. The app keeps the screen awake while it is active and restores the previous Android volume after the mode is turned off.

SECTION 08

LONGEVITY TEST

The **LONGEVITY TEST** runs the Cable Trigger through a set number of actuation cycles autonomously, using your current Pre-Trigger Delay, Exposure Time, and Trigger Engagement Amount settings. Use it to burn in a new cable release mechanism, verify reliability under repeated use, or simulate a long shoot.

Expand the LONGEVITY TEST row inside the CONFIGURATION card.

RUNNING A LONGEVITY TEST

1. Set the number of **CYCLES** using the +/− buttons or type directly into the counter field (1 to 9,999 cycles; initial count 10).
2. Tap **START TEST**. The device begins firing immediately, repeating the full trigger sequence for the requested number of cycles.
3. The progress display shows cycles remaining. The test continues even if you disconnect the phone.
4. Tap **STOP TEST** at any time to abort.

> The longevity test runs autonomously on the device — it is safe to disconnect the phone once it has started. The test uses your current delay, exposure, and engagement settings, so dial those in before starting.

SECTION 09

LINKING TO A LAZER EYE

The Cable Trigger can be triggered automatically by a **Lazer Eye** LiDAR sensor in two ways. Both are configured from the **LAZER EYE tab** of the SHP Asset Manager app.

DIRECT CABLE TRIGGER LINK (ESP-NOW)

The Lazer Eye sends a wireless trigger packet directly to the Cable Trigger over **ESP-NOW** every time the LiDAR detects a subject. Once paired, this link operates autonomously — no phone needs to be connected after setup. The Cable Trigger fires using whatever Pre-Trigger Delay, Exposure Time, and Trigger Engagement Amount settings are currently stored on it.

> Lazer Eye firmware **v2.9.4 or later** is required for confirmed-delivery status. Older firmware sends the packet but cannot confirm the Cable Trigger received it.

1. Connect the **Cable Trigger** to the app via BLE (the orange dot in the Cable Trigger tab header confirms the connection).
2. Open the **LAZER EYE tab** and connect the Lazer Eye TX by WiFi or BLE.
3. Expand **DIRECT CABLE TRIGGER LINK**. Tap **GET MAC** — the app reads the Cable Trigger's ESP-NOW MAC over BLE and displays it.
4. Tap **PAIR**. The Lazer Eye stores the Cable Trigger's MAC address and begins firing it on every accepted LiDAR detection.
5. Tap **TEST LINK** to send a test packet. Listen for a shutter click from the Cable Trigger. If delivery is confirmed you will see a green "Delivery confirmed" message; on older Lazer Eye firmware you will see a yellow "Sent (delivery not confirmed)" message.

Both devices remember the pairing after power cycling — you only need to pair once. To remove the link, expand DIRECT CABLE TRIGGER LINK and tap **UNPAIR**.

BLE-MEDIATED LINK

While both the Cable Trigger and Lazer Eye are connected to the app simultaneously, you can enable the **FIRE CABLE TRIGGER ON LIDAR EVENT** toggle in the **CABLE TRIGGER LINK** section of the Lazer Eye tab. When enabled, the app relays each accepted LiDAR trigger to the Cable Trigger over BLE in real time.

- Each relay uses the Cable Trigger's saved delay, exposure, and engagement settings.
- If the Cable Trigger is mid-sequence when an event arrives, that event is **skipped** — events are not queued.

› The link stops if either device disconnects. No pairing is stored; enable the toggle again after reconnecting.

CHOOSING THE RIGHT APPROACH

SCENARIO	RECOMMENDED LINK
Unattended setup — phone pocketed or left behind	**Direct Cable Trigger Link** — operates without a phone after pairing.
Phone already monitoring both devices simultaneously	**BLE-Mediated Link** — no extra pairing required; toggle on when needed.
Maximum reliability & confirmed delivery	**Direct Cable Trigger Link** with Lazer Eye firmware v2.9.4+ for delivery confirmation.

SECTION 10

TROUBLESHOOTING

Camera doesn't fire when TRIGGER SEQUENCE is tapped	Increase **Trigger Engagement Amount** — the servo may not be travelling far enough to press the plunger. Increase gradually and test until the shutter fires reliably. Also confirm the cable release is securely attached to the camera port.
Camera fires inconsistently	Increase **Exposure Time** gradually. You can run **Shutter Calibration** (TEST SHUTTER), save the measured profile, and tap APPLY to use its offset for the current connection; the profile does not apply automatically.
Servo strains or over-drives the cable release	Lower **Trigger Engagement Amount**. Find the minimum percentage that reliably fires the shutter — this reduces mechanical stress on the release and the camera's shutter port.
TEST SHUTTER shows no detection result or an error	Hold your phone close to the camera with the microphone facing the shutter and keep the area quiet. Allow microphone access for SHP Asset Manager in phone Settings. Confirm Trigger Engagement Amount is already high enough for the shutter to fire during the test.
Applying a calibration profile has no visible effect	The APPLY action sends a corrected Exposure Time for the current session only. Disconnect and reconnect to restore the original value, then APPLY again. If the correction is consistently needed, manually adjust **Exposure Time** in the CONFIGURATION card to match the corrected value.
Volume buttons don't trigger on Android	Confirm the remote trigger switch is on and the Cable Trigger screen is active. If another app has captured the hardware button events, close it and try again.
BLE clicker doesn't fire on iOS	Confirm the clicker is paired in iOS Settings → Bluetooth and shows as connected. Then confirm the remote trigger switch is on and the Cable Trigger screen is active.

Can't connect via Bluetooth	Toggle Bluetooth off and on and power-cycle the Cable Trigger. Allow Bluetooth access for SHP Asset Manager in phone Settings. On Android, allow nearby-device permission on Android 12+ or Location permission on Android 10–11. The scan list uses the device's current name, which can be changed in Settings.
Settings revert after reconnecting	Settings are written to the device immediately while connected. Ensure the device is connected (orange dot in header) when you make changes. The Shutter Calibration APPLY correction is session-only and must be re-applied each connection if needed permanently.
A firmware update is slow or failing	Use WiFi for a faster transfer. Open Settings → HARDWARE SETTINGS → FIRMWARE UPDATE (OTA), check for updates, and download the release before joining the device hotspot. Then use INSTALL VIA WIFI — INSTANT.
Direct link shows "sent (delivery not confirmed)" instead of "delivery confirmed"	Confirmed delivery requires **Lazer Eye firmware v2.9.4 or later**. Update the Lazer Eye via the Lazer Eye tab → Settings → CHECK FOR UPDATES. After updating, re-pair (UNPAIR then repeat the GET MAC → PAIR steps) and TEST LINK again.
Direct Cable Trigger Link set up but the Cable Trigger doesn't fire	Confirm the Lazer Eye TX is connected, ESP-NOW is enabled, and the Cable Trigger is powered. The Cable Trigger must be connected over BLE while you use GET MAC to pair. If TEST LINK fails, unpair and repeat the GET MAC → PAIR steps. Radio range varies with placement, walls, and interference.
BLE-mediated link fires inconsistently or skips events	Both the Cable Trigger and Lazer Eye must remain connected to the app throughout the session — the link stops if either disconnects. Events are **not queued**: if the Cable Trigger is mid-sequence when a LiDAR event arrives, that event is skipped. Reduce your LiDAR sensitivity or increase inter-detection interval to lower the event rate relative to the Cable Trigger's cycle time.
Direct link stops working after powering both devices off and back on	Open the Lazer Eye tab and check the DIRECT CABLE TRIGGER LINK status. If it shows NOT PAIRED, confirm ESP-NOW is enabled and repeat GET MAC → PAIR while the Cable Trigger is connected over BLE. Tap TEST LINK to verify delivery.

UPDATING FIRMWARE

1. Connect to the Cable Trigger in the app.
2. Open Settings → **HARDWARE SETTINGS** → **FIRMWARE UPDATE (OTA)** .
3. Tap **CHECK FOR UPDATES** , or use **PICK .BIN FILE** for firmware already on the phone.
4. Install over Bluetooth, or download the release first, join the Cable Trigger hotspot, and use **INSTALL VIA WIFI — INSTANT** .
5. Keep power connected and do not interrupt the transfer. The device reboots and the app attempts to reconnect after a successful update.
